

MAXILERATOR

Edge-Optimized AI Accelerator for CNN Inference

Team Members	Affiliation	Experience
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Project Information

Goal

To design and tape out a fully functional edge-optimized INT8 AI accelerator for CNN inference on the GF180MCU process node, capable of executing convolutional neural network layers using a weight-stationary systolic array architecture. The chip will demonstrate end-to-end CNN inference in real silicon, from weight loading to final classification output, verified against a software golden model.

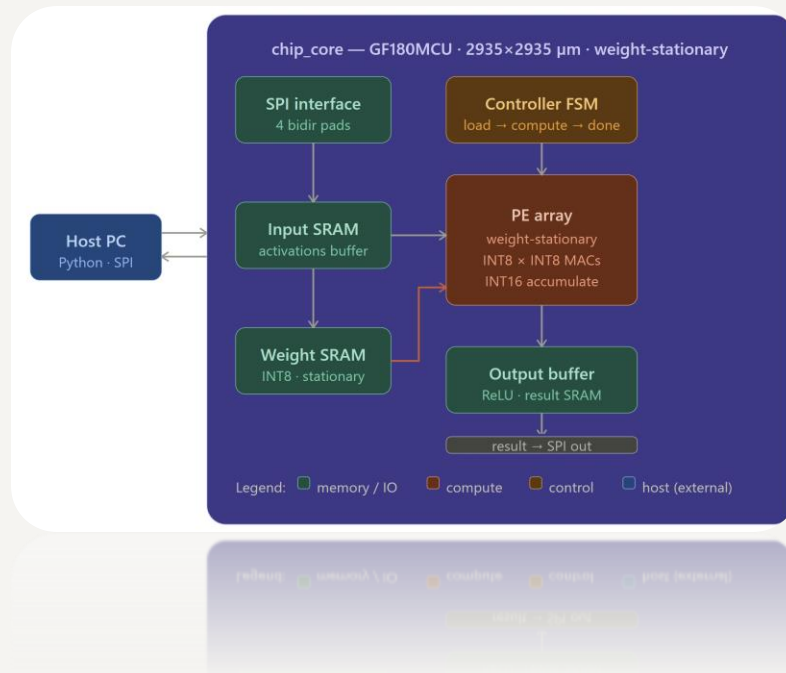
Application

The accelerator targets CNN-based image classification tasks, enabling efficient on-chip inference using pre-trained INT8 quantized weights without any external processor or memory.

Project Information (contd.)

Design

The design implements a weight-stationary systolic PE array of INT8 MAC units, where weights are preloaded into on-chip SRAM and held stationary while input activations are streamed through the array, producing accumulated partial sums at each processing element. The chip consists of five functional blocks — an SPI interface, input SRAM, weight SRAM, PE array with INT8 multiply-accumulate units, and an output buffer with ReLU activation — all integrated into a single chip core on the GF180MCU workshop slot die. The design targets a 25 MHz clock frequency and fits comfortably within the 2935×2935 μm dedicated die area.



Team Background

Academic Experience

Five-member international graduate team with strong digital hardware backgrounds, currently pursuing Master's degrees at leading technical universities — Technical University of Munich (TUM, Germany), Indian Institute of Technology Delhi (IIT Delhi, India), and National Taiwan University of Science and Technology (NTUST, Taiwan).

Work Experience

The team brings hands-on RTL design experience from academic projects and coursework, including prior tape out experience. Members have backgrounds in digital IC design, Verilog/SystemVerilog RTL development, and FPGA-based prototyping.

References

- S. Vishwakarma, Z. Khan, and J. J. Lehtomäki, "Impact of MAC unit design architectures and their applications in modern computing: An in-depth review," IEEE Access, vol. 14, pp. 24339–24360, 2026, doi: 10.1109/ACCESS.2026.3664183.
- S.-Y. Lee, M.-Y. Ku, S.-Y. Pan, and C.-C. Lin, "Reconfigurable and scalable artificial intelligence acceleration hardware architecture with RISC-V CNN coprocessor for real-time seizure detection," IEEE Access, vol. 13, pp. 31057–31068, 2025, doi: 10.1109/ACCESS.2025.3538781.
- S.-Y. Lee, Y.-W. Hung, Y.-T. Chang, C.-C. Lin, and G.-S. Shieh, "RISC-V CNN coprocessor for real-time epilepsy detection in wearable application," IEEE Trans. Biomed. Circuits Syst., vol. 15, no. 4, pp. 679-69, Aug. 2021.

Questions, Suggestions, Doubts?

- Does this design have a reasonable level of complexity for Track A, or should we scope it further, for example, limiting to a single conv layer rather than full multi-layer inference?
- For computing subsequent CNN layers, we plan to read the output back via the host PC and re-feed it as the next layer's input over SPI. Is this software-controlled iterative approach acceptable, or must the inter-layer feedback loop be implemented entirely in on-chip hardware?
- Which SystemVerilog language features are supported by the LibreLane slang frontend for GF180MCU, specifically, do parameterized structs, always_ff, and \$clog2 synthesize cleanly?
- What core utilization do you recommend to safely close timing at 25 MHz?